

Data Quality Issues and Resolutions Report

This report provides a summary of the data quality issues identified and addressed during the ETL process, specifically within the `clean_and_transform_data` function. The goal was to ensure the data loaded into the Snowflake data warehouse is accurate, consistent, and reliable for analysis.

1. Missing Values

Issues Identified:

- **products dataset:** Missing values were found in the `product_category_name`, `product_name_lenght`, `product_description_lenght`, `product_photos_qty`, `product_weight_g`, `product_length_cm`, `product_height_cm`, and `product_width_cm` columns.
- **orders dataset:** Missing values were present in the timestamp columns (`order_approved_at`, `order_delivered_carrier_date`, `order_delivered_customer_date`, `order_estimated_delivery_date`).
- **order_reviews dataset:** Missing values were found in the `review_comment_title` and `review_comment_message` columns.

Resolutions:

- **products dataset:** Rows with missing `product_category_name` were removed. Missing numerical values in product dimensions and weight were imputed using the median value of their respective columns.
- **orders dataset:** Timestamp columns were converted to datetime objects, and invalid date formats were coerced to NaT (Not a Time). Orders with inconsistent timestamps (e.g., approved date before purchase date) were removed. NaT values in timestamp columns were replaced with empty strings before loading to Snowflake.
- **order_reviews dataset:** Missing values in `review_comment_title` and `review_comment_message` were handled by converting them to empty strings after accent removal.

2. Data Type and Format Issues

Issues Identified:

- **Timestamp columns:** Dates in the orders and order_reviews datasets were not consistently in a datetime format suitable for direct loading into Snowflake's TIMESTAMP data type.
- **customer_zip_code_prefix and seller_zip_code_prefix:** These were integer types but needed to be treated as strings to preserve leading zeros if any.
- **geolocation_zip_code_prefix:** This was an integer type and needed to be treated as a string.

Resolutions:

- **Timestamp columns:** Converted timestamp columns in orders and order_reviews to pandas datetime objects using `pd.to_datetime` with `errors='coerce'`. Subsequently, these were formatted into a 'YYYY-MM-DD HH:MM:SS' string format using `.dt.strftime('%Y-%m-%d %H:%M:%S')` and `fillna("")` to handle NaT values before loading.
- **customer_zip_code_prefix and seller_zip_code_prefix:** Handled implicitly by pandas reading as objects, which is suitable for string representation in Snowflake.
- **geolocation_zip_code_prefix:** Explicitly converted to string type using `.astype(str)`.

3. Inconsistent Data Entry and Formatting

Issues Identified:

- **seller_city and geolocation_city:** Inconsistent casing, leading/trailing spaces, and accents were present.
- **payment_type:** The value "boleto" needed to be standardized to "bank_slip".
- **product_name_lenght and product_description_lenght:** Typographical errors in column names.

Resolutions:

- **seller_city and geolocation_city:** Converted to lowercase, stripped leading/trailing spaces, and removed accents using a custom lambda function with `unicodedata`. Manual corrections were applied to a few specific city names with known inconsistencies.
- **payment_type:** Replaced "boleto" with "bank_slip" using the `.replace()` method.
- **product_name_lenght and product_description_lenght:** Renamed the columns to `product_name_length` and `product_description_length` respectively using `.rename()`.

4. Duplicates

Issues Identified:

- **order_reviews dataset:** Duplicate reviews were found based on review_id.

Resolutions:

- **order_reviews dataset:** Duplicate rows were removed based on the review_id column using .drop_duplicates(subset=['review_id'], keep='first').

5. Invalid Data Values

Issues Identified:

- **products dataset:** Products with zero weight, length, height, or width were present.
- **fact_order_items dataset:** Orders with non-positive price values were found.

Resolutions:

- **products dataset:** Rows with zero values in product_weight_g, product_length_cm, product_height_cm, and product_width_cm were removed.
- **fact_order_items dataset:** Rows with price values less than or equal to zero were removed. Freight values less than zero were also removed.

6. Referential Integrity Issues

Issues Identified:

- **fact_order_items dataset:** There were records in fact_order_items that did not have corresponding entries in dim_orders, dim_products, or dim_sellers after cleaning these dimension tables.

Resolutions:

- **fact_order_items dataset:** Records in fact_order_items whose order_id, product_id, or seller_id were not present in the cleaned dim_orders, dim_products, or dim_sellers dataframes were removed to maintain referential integrity.

By implementing these cleaning and transformation steps, the data quality has been significantly improved, making the data warehouse a more reliable source for analytical purposes. Post-validation checks in Snowflake confirmed the successful resolution of most of these issues, with the exception of duplicates in the Dim_Geolocation table, which is expected due to multiple entries for the same zip code prefix, latitude, and longitude combination in the source data. An orphaned seller_id was also noted in the post-validation, which indicates a record in Fact_Order_Items that didn't have a matching seller in Dim_Sellers after cleaning.